



**VAAAL UNIVERSITY
OF TECHNOLOGY**
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FACULTY:

Appli

ed and Computer Sciences

DEPARTMENT:

Computer Sciences

SUBJECT: Business Analysis 3.1 / Financial

Information Systems 3.1 SUBJECT CODE: AIBUX3B /

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1. Introduction

This project is about developing a VUT Cafeteria Online Food Ordering System for the campus that makes it easier for students, staff, and visitors to order food at the cafeteria online, track their orders, and collect their food when it's ready. Instead of standing in long queues or dealing with delays that waste valuable time for students or staff members in the university. Users will be able to register and login as students or staff, browse available menus, place their orders, and make payments directly from their devices. This approach can help reduce congestion at the cafeteria and improve the overall efficiency of food services on campus.

The main objective of this project is to create a simple, reliable, and user-friendly platform that meets the needs of both customers and food service providers.

2. PROJECT SCOPE

2.1 Project Overview

The project scope defines the features and deliverables included in the development of the VUT Cafeteria Online Food Ordering System. It also outlines the project boundaries by specifying what is included and excluded, ensuring that the project remains focused and preventing scope creep.

The system aims to improve the efficiency of campus cafeteria services by allowing students to place food orders online rather than waiting in long queues. Through the system, students will be able to browse menus, place orders, and monitor the status of their orders. Cafeteria vendors will manage incoming orders through a vendor dashboard.

The system is expected to improve service efficiency, reduce waiting times, and provide a more convenient food ordering experience for both students and cafeteria vendors.

2.2 Project Objectives

The main objectives of this project are to:

- Develop a web-based food ordering system for the VUT campus cafeteria.
- Improve the efficiency of cafeteria services by reducing queues and waiting times.
- Provide a convenient platform for students to order food before arriving at the cafeteria.
- Enable cafeteria vendors to manage and process orders digitally.

2.3 In Scope Deliverables

The following deliverables are included in the scope of this project.

Customer / Student Deliverables

- Student registration and login functionality
- Menu browsing interface displaying available cafeteria food items
- Online food order placement functionality
- Order confirmation notification after placing an order
- Real-time order tracking showing order progress
- Order history allowing students to view previous orders

Food Service Provider (Vendor) Deliverables

- Vendor login and authentication system
- Vendor dashboard for viewing incoming orders
- Order management tools for accepting, preparing, and completing orders
- Order status update feature to reflect order progress
- Basic reporting interface for viewing daily orders

System Deliverables

- A centralized database storing user accounts, menu items, and order information
- A web-based platform accessible via standard web browsers
- Administrative functionality for managing menu items and system data
- Order tracking mechanism that updates order status for customers

2.4 Out of Scope

The following features will not be included in the VUT Cafeteria Online Food Ordering System:

- Integration with external restaurants or third-party food delivery services
- GPS-based delivery tracking
- Mobile application development (the system will only be web-based)
- Residence or student accommodation delivery services

3. Work Breakdown Structure (WBS) for Online Food Ordering System

1.1 Project Initiation

- 1.1.1 Identify stakeholders
- 1.1.2 Conduct stakeholder meetings
- 1.1.3 Define project objectives
- 1.1.4 Develop project charter
- 1.1.5 Approve project charter

1.2 Project Planning

- 1.2.1 Define project scope
- 1.2.2 Develop detailed WBS

- 1.2.3 Create project schedule
- 1.2.4 Estimate project costs
- 1.2.5 Risk management planning
- 1.2.6 Resource planning

1.3 System Analysis

- 1.3.1 Gather user requirements
 - 1.3.1.1 Interview students
 - 1.3.1.2 Interview vendors
 - 1.3.1.3 Collect admin requirements
- 1.3.2 Analyze requirements
- 1.3.3 Create system requirement specification (SRS)
- 1.3.4 Review and approve requirements

1.4 System Design

- 1.4.1 Design system architecture
- 1.4.2 Database design
 - 1.4.2.1 Define entities (Users, Orders, Menu)
 - 1.4.2.2 Create ER diagram
- 1.4.3 User interface design
 - 1.4.3.1 Student interface design
 - 1.4.3.2 Vendor interface design
 - 1.4.3.3 Admin dashboard design
- 1.4.4 System design review

1.5 System Development

- 1.5.1 Front-end development
 - 1.5.1.1 Login & registration
 - 1.5.1.2 Menu browsing
 - 1.5.1.3 Cart & checkout
- 1.5.2 Back-end development
 - 1.5.2.1 Database implementation
 - 1.5.2.2 Order processing logic
 - 1.5.2.3 Payment integration
- 1.5.3 Vendor dashboard development
 - 1.5.3.1 Menu management
 - 1.5.3.2 Order management
- 1.5.4 Integration of system components

1.6 Testing

1.6.1 Unit testing

1.6.2 Integration testing

1.6.3 System testing

1.6.4 User Acceptance Testing (UAT)

1.6.5 Bug fixing

1.7 Deployment

1.7.1 Prepare deployment environment

1.7.2 Deploy system to server

1.7.3 System configuration

1.7.4 User training

1.7.4.1 Train students

1.7.4.2 Train vendors

1.7.5 Go-live

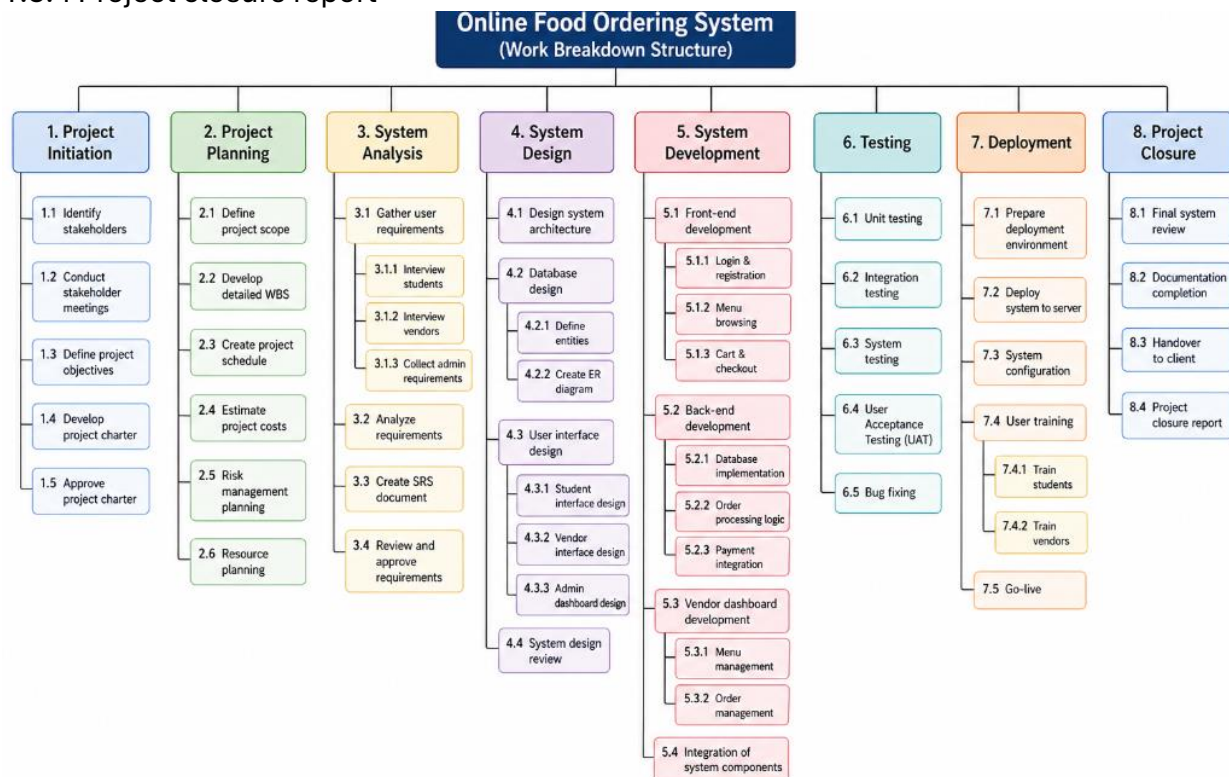
1.8 Project Closure

1.8.1 Final system review

1.8.2 Documentation completion

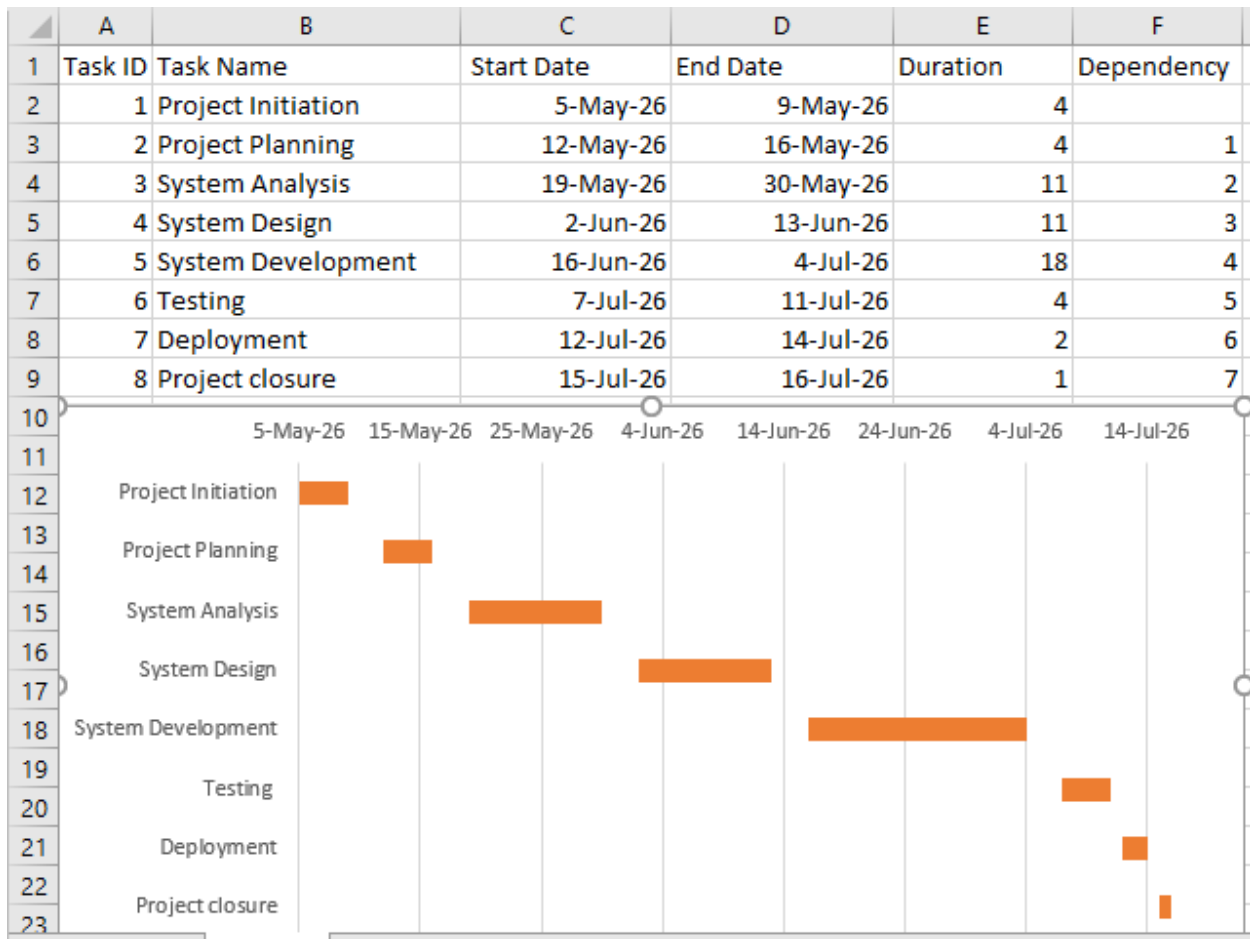
1.8.3 Handover to client

1.8.4 Project closure report



4. Project Schedule and Gantt Chart

The project schedule outlines the sequence of activities, their durations, and dependencies to successfully complete the VUT Online Food Ordering System. The schedule ensures that tasks are completed in the correct order and within the allocated timeframe. The total estimated project duration is 11 weeks.



5. RESOURCE PLAN

The resource plan outlines the human, hardware, and software resources required to develop and implement the VUT Cafeteria Online Food Ordering System across all project phases.

5.1 Human Resources (Mapped to Tasks & Constraints)

- Project Managers → Supervise project phases, ensure deadlines are met. *Constraint: limited to 2 managers due to budget.*
- Software Developers → Build front-end (ordering interface) and back-end (payment, menu management). *Constraint: must deliver within 3 months.*
- Database Administrator → Design and maintain MySQL database for menus, orders, and user accounts. *Constraint: ensure 99.9% uptime.*

- UI/UX Designer → Create student-friendly interface for mobile and desktop. *Constraint: must comply with accessibility standards.*
- System Analyst → Gather requirements from cafeteria staff and students, define specifications. *Constraint: must finalize before development starts.*
- Testers (QA Team) → Conduct unit, integration, and system testing. *Constraint: testing cycles limited to 2 weeks per sprint.*
- Network Administrator → Maintain connectivity, configure servers/cloud hosting. *Constraint: ensure redundancy and backup systems.*

5.2 Hardware and Technology Equipment

- Computers/laptops → Development and administration tasks
- Smartphones → Mobile compatibility testing
- Servers/cloud hosting → System deployment
- Internet access → Campus Wi-Fi and LAN
- Backup storage → External drives/cloud storage

5.3 Software and Other Resources

- IDEs: NetBeans, VS Code
- Database: MySQL
- Web stack: HTML, CSS, JavaScript
- Version control: Git
- Documentation: MS Word, PDF tools
- Testing/debugging tools
- Payment integration APIs

6. QUALITY PLAN

The quality plan ensures that the system meets required standards of performance, usability, and reliability.

6.1 Quality Standards

- Users can register, log in, browse menus, and place orders successfully
- Simple, intuitive interface
- Fast page load and efficient order processing
- Minimal downtime (target: < 0.5%)
- Secure handling of user data
- Cross-device and cross-browser compatibility
- Accurate data (prices, orders, transactions)

6.2 Quality Measurement (with KPIs)

- Page Load Time → Target < 3 seconds

- Order Processing Time → Target < 5 seconds
- System Uptime → ≥ 99.5%
- Error Rate → < 2% of transactions
- Bug Tracking → ≤ 5 critical bugs per release
- Successful vs Failed Transactions → ≥ 95% success rate
- User Feedback Score → ≥ 80% satisfaction rating

6.3 Quality Monitoring and Control

- Continuous testing after deployment
- Performance monitoring tools (track response time, uptime, error rate)
- User support and feedback channels
- Regular updates and maintenance
- Security monitoring and vulnerability check

7. Risk Management Plan

7.1 Risk monitoring

- Risks will be reviewed weekly during project meetings.
- A risk register will be maintained, including current scores, owners, and mitigation status.
- Escalation procedures are in place: risks with score
- require senior management notification within 24 hours.
- Owners are responsible for updating mitigation progress and recalculating scores after review.

Risk	Impact	Probability	Risk
Vendor resistance to adoption	High(3)	Medium (2)	6
Low student adoption	High(3)	Medium (2)	6
Payment integration delays	Medium (2)	Medium (2)	4
Network infrastructure issues	Medium (2)	Low (1)	2
Scope creep	Medium (2)	High (3)	6

8. COMMUNICATION PLAN

Communication Channels

- The use of e-mail for formal communications and documentation
- WhatsApp for daily collaboration and quick updates
- Google docs for task tracking, technical discussions and for document management and sharing

Activity	Frequency	Audience	Purpose
Daily Standup	Daily	Project Team	Progress updates
Sprint Planning	Weekly	Development Team	Task allocation

Stakeholder Meeting	Monthly	Sponsors, Vendors	Status reporting
Progress Report	Weekly	Project Sponsor	Written updates
User Feedback Session	Weekly	Students	Requirements validation

9. Project Budget

9.1 Cost Estimation Approach

The budget was created using a bottom-up approach and both direct costs and indirect costs were considered. A contingency amount was also added to cover unexpected costs.

9.2 Estimated Project Costs

Category	Description	Cost (In Rands)
Personnel Costs		
Business Analyst	Gathering and analyzing requirements	120 000
System Designer	Designing system and user interface	100 000
Developer	System development (front-end & back-end)	360 000
Tester	Testing the system	80 000
Project Manager	Managing and monitoring the project	150 000
SUBTOTAL		810 000
Technology Costs		
Web Hosting	Hosting the system online	40 000
Domain Name	Website registration	3 000
Development tools	Software and tools	50 000
Database Setup	Database configuration	35 000
SUBTOTAL		128 000
Training & Deployment	Training users and system setup	40 000
Miscellaneous Costs	Admin , communication , documentation	25 000

Contingency (10%)	Extra costs for unexpected issues	112 300
TOTAL BUDGET		R1 115 300

9.3 Cost Baseline

The estimated R1 115 300 will be used as the cost baseline to measure and control spending during the project.

9.4 Budget Control

To ensure the project stays within budget:

- Actual costs will be checked against planned costs.
- Any overspending will be identified early.
- Changes that affect the budget must be approved.
- Regular monitoring will be conducted throughout the project.

9.5 Assumptions

The following assumptions were made:

- The project will take about 11 weeks to complete.
- Most tools used will be affordable or free.
- The team is small, and some members may handle more than one role.
- No expensive hardware is required.

10. Conclusion

In conclusion, this project plan outlines the development of the VUT Cafeteria Online Food Ordering System, which aims to improve the efficiency of food services on campus. The system is designed to reduce long queues, save time for students and staff, and provide a more convenient way of ordering food online.

The project scope clearly defines what will be included in the system, such as user registration, menu browsing, online ordering, and order tracking. The Work Breakdown Structure (WBS) and project schedule provide a clear structure of the tasks and timelines needed to complete the project within the estimated 11 weeks.

In addition, the communication plan ensures that all stakeholders are kept informed throughout the project, while the project budget provides a realistic estimate of the financial resources required. Proper cost management and monitoring will help ensure that the project stays within budget. Overall, this project plan provides a clear roadmap for successfully developing and implementing the system. If all phases are properly executed, the system will meet its objectives and improve the overall cafeteria experience at VUT.